Q1. (a) Green beans contain vitamin C.

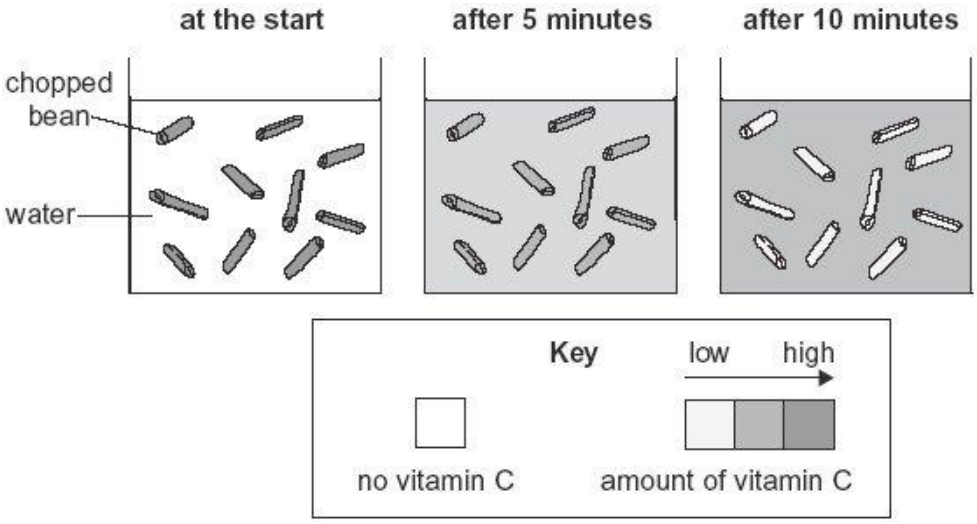


Which other food is a good source of vitamin C?
Tick the correct box.

cheese	chicken	eggs	oranges
☐	☐	☐	☐

1 mark

(b) The amount of vitamin C changes in the beans and in the water as the beans are cooked.
The shading shows how it changes.



Use the diagram. How does the **amount of vitamin C** in the beans and in the water change as the beans are cooked? Tick one box in each row.

amount of vitamin C	increases	decreases	stays the same
in the beans			
in the water			

1 mark

(c) Cheese is a source of calcium.

Why do we need calcium?

.....

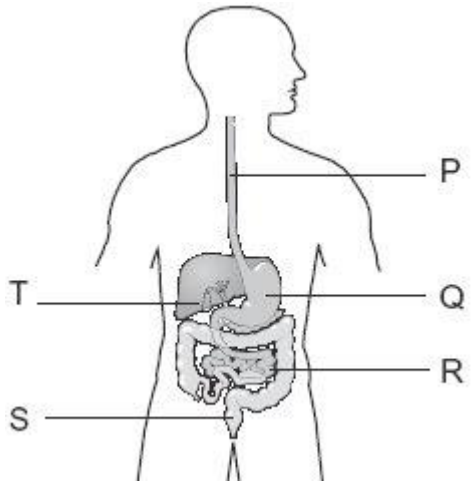
1 mark

(d) Draw a line from each nutrient to a good source of that nutrient in our diet.

nutrient	source of nutrient
starch	lean chicken meat
fat	jam
protein	pasta
sugar	margarine

2 marks

(e) The diagram shows part of the human digestive system.



(i) Write the letter which labels the small intestine.

.....

1 mark

(ii) Write the letter which labels the stomach.

.....

1 mark
maximum 7 marks

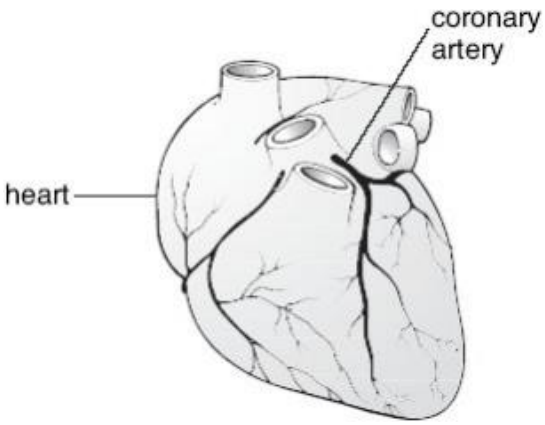
Q2. (a) Carbon monoxide, nicotine and tar get into the lungs when a person smokes.

Draw a line from each substance to the effect of the substance on the body.
Draw only **three** lines.

substance	effect of the substance
carbon monoxide	causes addiction to smoking
nicotine	causes influenza (flu)
tar	causes lung cancer
	causes red blood cells to carry less oxygen

3 marks

(b) The coronary arteries carry blood to the heart muscle.
The drawing below shows the heart and coronary arteries.



(i) Diagram 1 shows a section through a coronary artery.

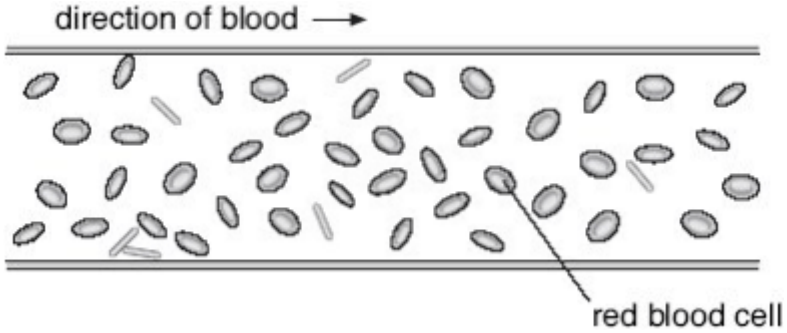


diagram 1

Smoking can cause damage to the coronary artery.
Diagram 2 shows a section through part of a damaged artery.

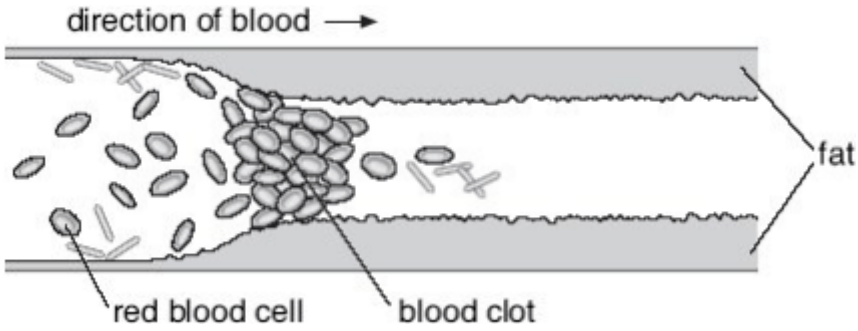


diagram 2

not to scale

Look at diagram 2. A blood clot has formed.

Give **one** other change in the coronary artery.

.....

.....

1 mark

(ii) Respiration takes place in the muscle cells of the heart.

Explain why a blood clot in the coronary artery prevents these cells respiring normally.

.....

.....

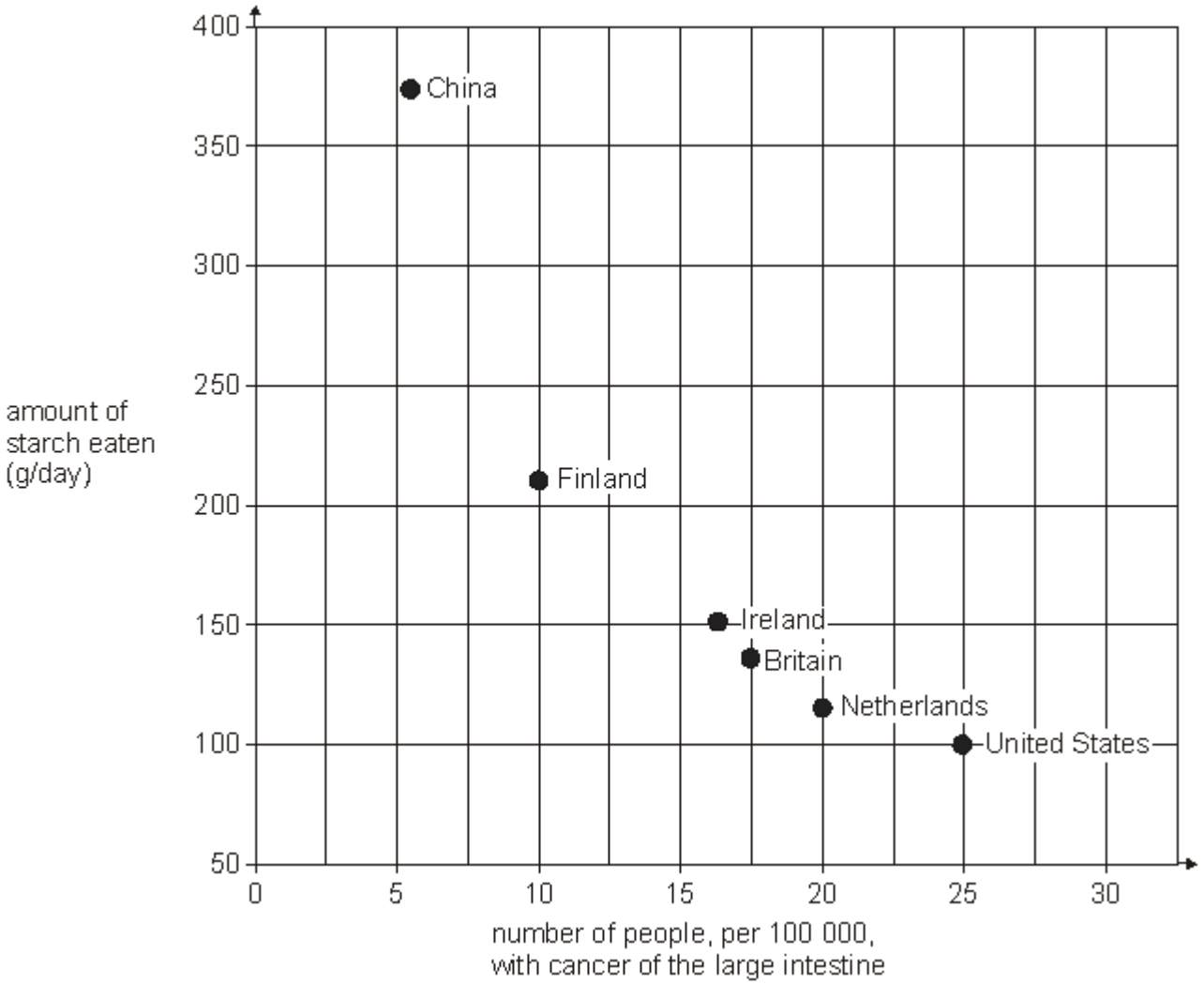
.....

.....

2 marks
maximum 6 marks

Q3. People in different countries eat different amounts of starch.
A scientist compared the amount of starch that people ate with the number of people with cancer of the large intestine.

The scatter graph below shows her results.



Look at the scatter graph.

(a) (i) Which country had the greatest proportion of people with cancer of the large intestine?

.....

1 mark

(ii) What conclusion could you come to about the effect of eating starch on getting cancer of the large intestine?

.....

.....

1 mark

(b) (i) Starch is a carbohydrate.

Which **two** of the following foods are good sources of starch?
Tick the **two** correct boxes.

bread	☐	cheese	☐
chicken	☐	tomatoes	☐
fish	☐	pasta	☐

2 marks

(ii) What other type of nutrient, needed as part of a balanced diet, keeps the intestine working well and prevents constipation?
Tick the correct box.

fat	fibre	minerals	protein	vitamins
☐	☐	☐	☐	☐

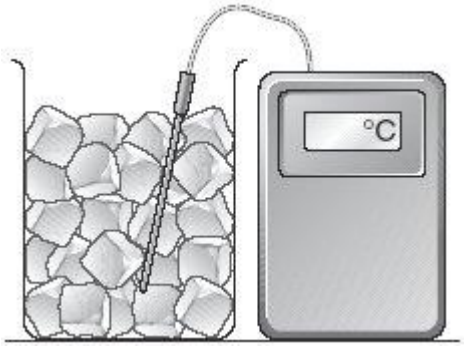
1 mark
maximum 5 marks

Q4. (a) Draw a line from each change of state to the correct name.
Draw only **four** lines.

change of state	name
solid to liquid	evaporating
liquid to gas	melting
gas to liquid	condensing
liquid to solid	freezing

3 marks

(b) Kate made some ice cubes from pure water.
She used a sensor to measure the temperature of the ice.



What temperature will the sensor show when the ice is melting?

..... °C

1 mark

(c) Kate made some more ice cubes from salt solutions. She used a different amount of salt in each ice cube.

The table shows the temperature at which the ice cubes melted.

mass of salt in each ice cube (g)	temperature ice cube melted (°C)
5	−4
10	−8
15	−11
20	−15

Look at the table above.
As the mass of salt increased, what happened to the temperature at which the ice cube melted?

.....

1 mark

(d) In very cold weather a mixture of salt and sand is spread on roads.

Why are salt **and** sand used?
Tick the **two** correct boxes.

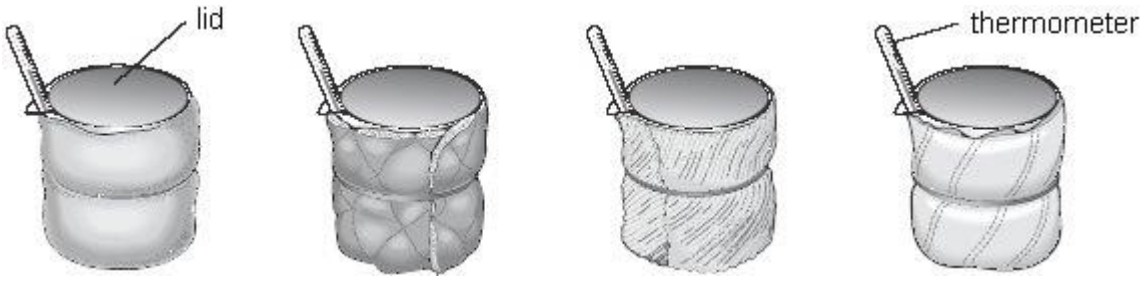
Salt makes the roads white.	☐	Sand dissolves in water.	☐
Salt makes water freeze.	☐	Sand increases friction between car tyres and the road.	☐
Salt makes ice melt.	☐	Sand makes water freeze.	☐

2 marks

maximum 7 marks

Q5. A company has made a new material called 'Wellwarm'. They want to use 'Wellwarm' to make coats.

(a) A scientist tested 'Wellwarm' to see how well it insulated a beaker of hot water.
She tested 'Wellwarm' and three other materials as shown below.



material A **material B** **material C** **material D**

She wrapped each beaker in a different material.
She recorded the temperature at the start and 20 minutes later.

(i) What was the independent variable that the scientist **changed**?

.....

1 mark

(ii) What was the dependent variable that the scientist **measured** during the investigation?

.....

1 mark

(b) The results of the investigation are shown below.

time (minutes)	temperature of water (°C) wrapped in			
	material A	material B	material C	material D
0	60	60	60	60
20	34	40	38	36

(i) The scientist said that the 'Wellwarm' material is the best insulator.
Which material was 'Wellwarm'? Use the results to help you.
Tick the correct box.

A ☐ B ☐ C ☐ D ☐

1 mark

(ii) Use the evidence in the results table to explain your choice.

.....

.....

1 mark

(c) The company made a coat from each of the four materials they tested.



A person tested the different coats by wearing each one in a cold room.
He measured the temperature inside each coat for 30 minutes.

Write down two **other** variables that should be controlled to make this a fair test.

1. .....

1 mark

2. .....

1 mark

(d) Write down one thing the scientists should do to make sure the person testing the coats is safe.

.....

1 mark

(e) Suggest **one** advantage of using a temperature sensor and data logger instead of a thermometer in this experiment.

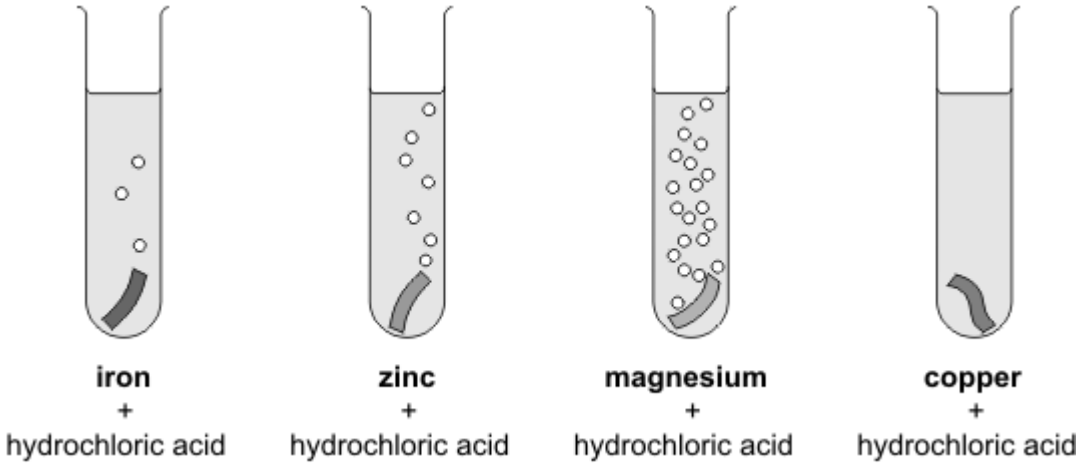
.....

.....

1 mark
maximum 8 marks

Q6. (a) Ruth put a piece of a different metal in each of four test tubes.

She poured 10 cm^3 of hydrochloric acid onto each metal.



Look at the diagrams above.

(i) How do these show if a metal reacts with the acid?

.....

1 mark

(ii) **On the lines below**, put the **four** metals in the order of how strongly they react with the acid.

most reactive .....

.....

.....

least reactive .....

1 mark

(b) Choose the name of a metal from the box below to answer each question.

copper	**iron**	**magnesium**	**zinc**

(i) Which metal from the box is used for electrical wires?

.....

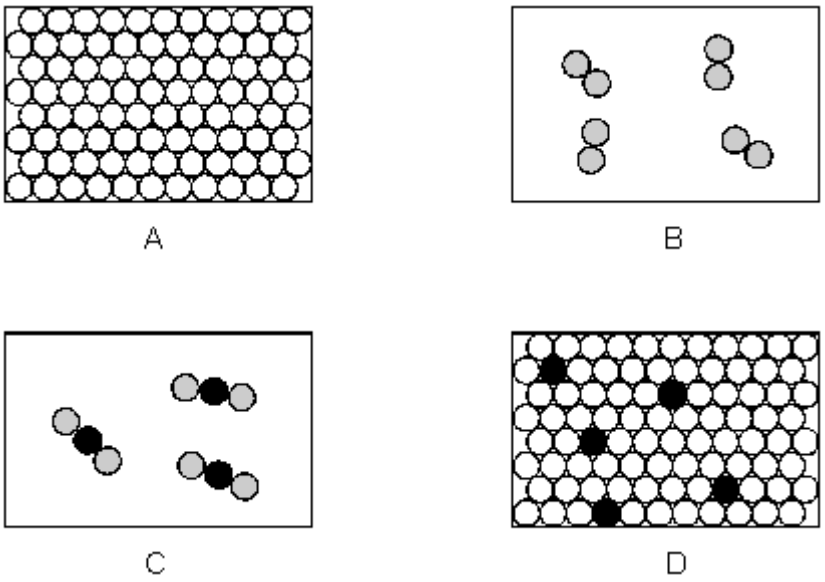
1 mark

(ii) Which metal from the box goes rusty?

.....

1 mark
maximum 4 marks

Q7. The diagrams represent the arrangement of atoms or molecules in four different substances, A, B, C and D.



not to scale

Each of the circles, ○, ◯ and ● represents an atom of a different element.

(a) (i) Which substance is a compound?

……

1 mark

(ii) Which substance is a mixture?

……

1 mark

(iii) Which **two** substances are elements?

…… and ……

1 mark

(iv) Which **two** substances could be good thermal conductors?

…… and ……

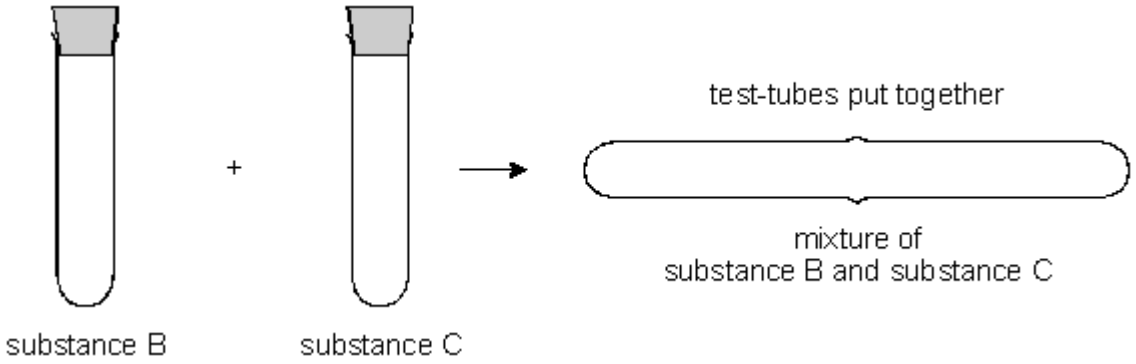
1 mark

(v) Which substance could be carbon dioxide?

……

1 mark

(b) The following experiment was set up. Test-tubes containing substances B and C were placed together as shown. The substances did **not** react.
They were left for five minutes.

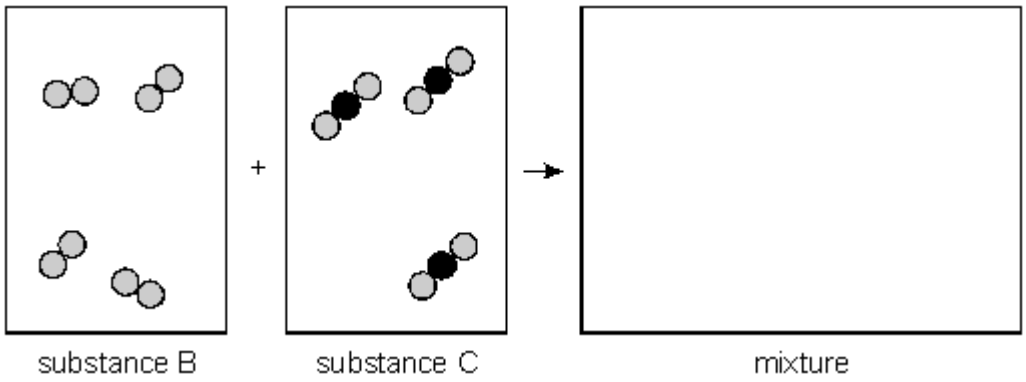


(i) How many molecules are there in the mixture compared to the total number in substances B and C?

……

1 mark

(ii) Complete the diagram which is a model of this experiment.



1 mark
Maximum 7 marks

Mark schemes

Q1.

(a) oranges ✓

if more than one box is ticked, award no mark

1 (L3)

(b)

amount of vitamin C	increases	decreases	stays the same
in the beans		✓	
in the water	✓		

***both** ticks are required for the mark*

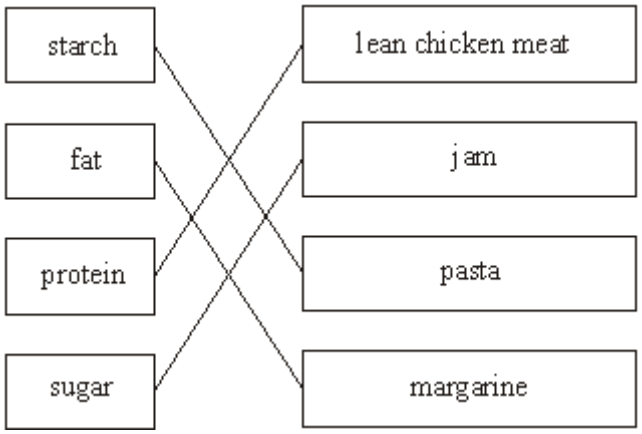
1 (L3)

(c) for strong **or** hard teeth **or** bones

accept 'to keep the skeleton strong'
*accept 'for (healthy) bones **or** teeth'*
*'for toe **or** finger nails' is insufficient*

1 (L4)

(d)



***all four** lines are required for two marks*
*any **two or three** lines are required for one mark*
if more than one line is drawn from any nutrient, do not give credit for that nutrient

2 (L3)

(e) (i) R

1 (L4)

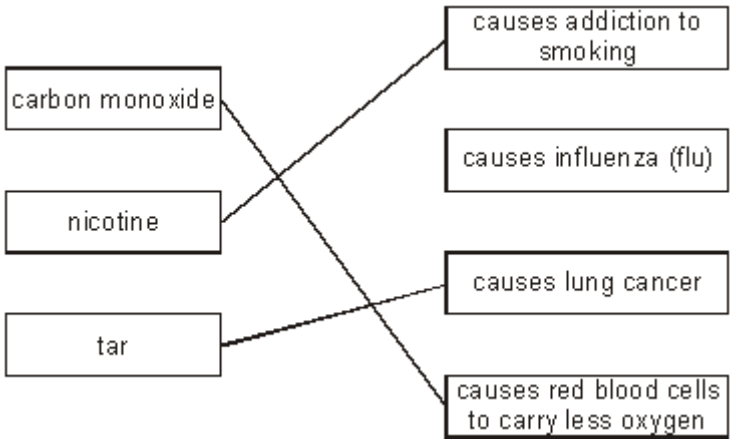
(ii) Q

1 (L4)

[7]

Q2.

(a)



if more than one line is drawn from any substance, award no mark for that substance

3 (L5)

(b) (i) any **one** from

- fat has been deposited
  accept 'fat'
- the artery has become narrower
  accept 'it is blocked'

1 (L5)

(ii) any **two** from

- less oxygen to the cells or heart muscle
  accept 'no oxygen'
  *do **not** accept 'less air to the cells'*
- less glucose to the cells **or** heart muscle
  accept 'no glucose'
- less blood to the cells **or** heart muscle
  accept 'it stops the circulation of blood'
  accept 'blood cannot flow through'
  accept 'it stops blood getting to the heart'
  'nothing can get through' is insufficient
  accept 'there is a build-up of carbon dioxide'
  accept for two marks 'they get less oxygenated blood'

2 (L6)

[6]

Q3.

(a) (i) United States

*accept 'US' **or** 'USA'*
accept 'America'
accept lower case letters for 'US'

1 (L5)

(ii) the less starch eaten, the more people get cancer

accept the converse

*do **not** accept 'people with cancer eat less starch'*

accept 'starch helps to prevent cancer'
accept 'you should eat a lot of starch'
'eat starch' is insufficient

1 (L5)

(b) (i) • bread ✓

• pasta ✓

if more than two boxes are ticked,
deduct one mark for each incorrect tick
minimum mark zero

1 (L6)

(ii) fibre ✓

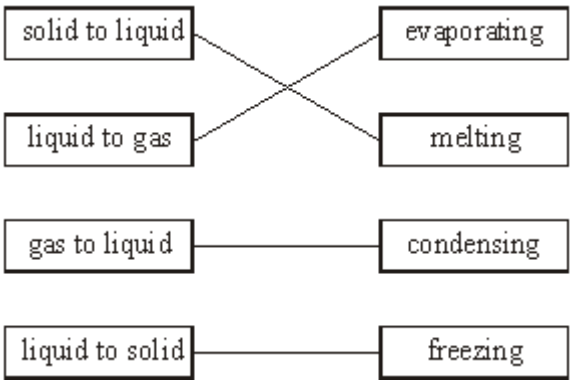
if more than one box is ticked, award no mark

1 (L6)

[5]

Q4.

(a) •



*award three marks for **all four** correct lines*
*award two marks for any **three** correct lines*
*award one mark for any **two** correct lines*

if more than one line is drawn from any change of state,
do not credit that change of state

3 (L3)

(b) • 0 °C

accept 'zero'

*do **not** accept 'nothing'*

1 (L4)

(c) • it decreased

accept 'it got colder'

'it dropped to below 0°C' is insufficient
any references to time are insufficient

1 (L3)

(d) • Sand increases friction between car tyres and the road. ✓

• Salt makes ice melt. ✓

if more than two boxes are ticked, deduct a mark for each incorrect box
minimum of zero

2 (L4)

[7]

Q5.

(a) (i) (the type of) material

accept 'fabric'

1 (L5)

(ii) the temperature after 20 minutes

accept 'temperature'
'temperature at start' is insufficient
accept 'temperature change'
accept 'how hot the water is'
'heat' is insufficient

1 (L5)

(b) (i) B ✓

if more than one box is ticked, award no mark

1 (L5)

(ii) beaker B was warmer (at the end than the others)

accept the converse
accept 'the temperature was higher'
accept 'it dropped the least'
accept 'it lost the least heat'
'B was 40°C at the end' is insufficient

1 (L5)

(c) any **two** from

• temperature in cold room

'temperature' is insufficient
'the person' or 'the time' is insufficient
accept 'his temperature before he went in'

• style or size of coat

accept 'amount of material'

• the clothes he wears for each test

accept '(same) underclothes'

• level of activity

accept 'he had eaten the same food before each test'

2 (L5)

(d) • do not let the volunteer's body temperature go down too far

accept 'do not let him or the room get too cold'

accept 'have a doctor nearby'
accept 'monitor or check him'
accept 'measure his heart rate or breathing rate'
accept 'wear gloves' or 'keep his head warm'
accept 'make sure the volunteer is healthy'
or 'is not allergic to the material'

1 (L5)

(e) any **one** from

- monitor the temperature from outside the room
  accept 'remote sensing'
  accept 'you have to open the coat to read the thermometer'
- it gives you a continuous record (of the temperature)
  accept 'you do not have to write down the results'
- it is more accurate or precise
  accept 'it eliminates human error'
  'accurate' is insufficient
  accept 'the experiment is more reliable'
  'more reliable' is insufficient

1 (L5)

[8]

Q6.

(a) (i) any **one** from

- bubbles
- fizzing
  accept 'effervescence'
- gas is given off
  *'metal goes into solution **or** turns into a salt'*
  and 'there would be a rise in temperature'
  are insufficient answers as they are
  ***not** shown in the drawings*

1 (L3)

(ii)
- magnesium
  accept 'Mg'
- zinc
  accept 'Zn'
- iron
  accept 'Fe'
- copper

accept 'Cu'

answers must be in the correct order

***all four** answers are required for the mark*

1 (L4)

(b) (i) • copper

accept 'Cu'

1 (L3)

(ii) • iron

accept 'Fe'

1 (L4)

[4]

Q7.

(a) (i) C

1 (L7)

(ii) D

1 (L7)

(iii) A and B

answers may be in either order

***both** answers are required for the mark*

1 (L7)

(iv) A and D

answers may be in either order

***both** answers are required for the mark*

1 (L7)

(v) C

1 (L7)

(b) (i) the same

accept 'seven'

1 (L7)

(ii) a random, mixed arrangement of both types of molecule should be drawn with the molecules not touching each other

1 (L7)

[7]